



Part I

Lab Assignment-6 (Introduction to Object-Oriented Programming)

Department: CSE
(22CPP209)

Subject: OOAD Lab

Objective: Understanding *Operator overloading* for binary operators.

Instruction: Students are advised to explore the code behavior by varying the implementation as member and non-member function.

Question 1. Create a class named **Complex** that is represented with two private data members `real` and `imaginary` in the form `real + j*imaginary`. Implement a constructor that initializes the real and imaginary parts of the complex number. Provide a default constructor that initializes both real and imaginary parts to 0.

- a) Overload the **addition** and **subtraction** operator to perform these operations on two complex numbers as in algebra.
- b) Overload the **multiplication** operator to enable multiplication of two complex numbers as in algebra.
- c) Modify the class to enable input and output of complex numbers via overloaded `>>` and `<<` operators, respectively.

Question 2: Create a class **RationalNumber** (fractions) with these capabilities:

- a) Create a constructor that prevents a 0 denominator in a fraction, reduces or simplifies fractions that are not in reduced form and avoids negative denominators.
- b) Overload the addition, subtraction, multiplication and division operators for this class.
- c) Overload the relational and equality operators for this class.